

Arrays and strings ★ Golden

One college period · 40–45 minutes · Programming using C

What you will learn

- Declare, initialise and traverse 1-D and 2-D arrays
- Explain string terminator '\0' and strlen, strcpy, strcat, strcmp
- Predict output of num[3] style MCQs

Array. A collection of elements of the same data type stored in contiguous memory, accessed by index starting at 0.

```
int num[] = {3, 5, 7, 10, 20};
printf("%d", num[3]);    /* 10 - index 3 is the fourth element */
```

```
int a[5], i, sum = 0;
for (i = 0; i < 5; i++) {
    scanf("%d", &a[i]);
    sum += a[i];
}
printf("Average = %.2f", sum / 5.0);
```

Two-dimensional array

```
int m[3][3], i, j;
for (i = 0; i < 3; i++)
    for (j = 0; j < 3; j++)
        scanf("%d", &m[i][j]);
```

Strings

A string is a char array ending with '\0'. char name[20] = "Ali"; /* 'A','l','i','\0' */

Function	Header	Use
strlen(s)	string.h	length without '\0'
strcpy(d,s)	string.h	copy s into d
strcat(d,s)	string.h	join s onto d
strcmp(a,b)	string.h	0 if equal, <0 or >0 if not

```
char s[30];
int i, n;
gets(s);                /* exam style */
n = strlen(s);
for (i = n - 1; i ≥ 0; i--)
    printf("%c", s[i]);  /* reverse name */
```

HOW BIEK ASKS THIS

2025: What is an array? Declare and initialise. Output of num[3] with {3,5,7,10,20} is 10 — not 7. Students forget index 0.

COMMON BOARD MISTAKE

scanf("%s", name) — no & for arrays because the name is already an address. Overrunning the array size crashes the program.

اردو میں سمجھیں

Array ایک ہی قسم کی کئی قیمتیں، انڈیکس 0 سے۔ `num[3]` چوتھا عنصر ہے۔
سٹرنگ حروف کی صف ہے جو `\0` پر ختم ہوتی ہے۔ `strlen` لمبائی، `strcpy` نقل، `strcat` جوڑ، `strcmp` موازنہ۔

CHECK YOURSELF

1. Index of the first element?

2. `num[]={3,5,7,10,20}; num[3]?`

Answers: 1. 0 · 2. 10

MCQs

1. Array index in C starts from:

- a. 1
- b. 0
- c. -1
- d. 2

2. A string in C ends with:

- a. `\n`
- b. `\0`
- c. `\t`
- d. NULL pointer only

Key: 1=b, 2=b

Short questions (3 marks each)

1. What is an array? Declare and initialise an integer array.
2. `int num[]={3,5,7,10,20};` what does `printf("%d", num[3])` print?
3. Write a program to reverse an inputted name using `strlen()`.
4. Name four string functions.

Model short answers

1. Same-type contiguous list. `int a[5]={1,2,3,4,5};` or `int a[]={10,20};`
2. 10.
3. Read string, loop from `strlen-1` down to 0 printing characters.
4. `strlen`, `strcpy`, `strcat`, `strcmp` (plus `strncpy` etc. if space).

Long questions (board Section C)

1. Explain array and its types with examples. Write a program to add two 3×3 matrices.